

Task 1 – Reconnaissance and Vulnerability Scanning

Objective

The objective of this task is to perform active reconnaissance and vulnerability scanning on a safe test system using network scanning and vulnerability assessment tools.

Tools Used

- Nmap
 - Nikto
 - Kali Linux
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Target Information

- Target Host: scanme.nmap.org
 - Target IP Address: 45.33.32.156
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Introduction

Reconnaissance and vulnerability scanning are important phases in cybersecurity assessments. These techniques help identify open ports, running services, and potential security weaknesses in a system.

In this task, Nmap was used for port and service enumeration, while Nikto was used to identify web server vulnerabilities and misconfigurations.

Step 1 – Nmap Scan

Command Used

```
nmap -sV -T4 scanme.nmap.org
```

Purpose

The purpose of this scan was to:

- Identify open ports
- Detect running services
- Determine service versions

Scan Findings

Port	State	Service	Version
22/tcp	Open	SSH	OpenSSH

Port	State	Service	Version
80/tcp	Open	HTTP	Apache HTTP Server
9929/tcp	Open	Nping Echo	Nping Echo Service
31337/tcp	Open	tcpwrapped	Unknown

Results

```
# Nmap 7.98 scan initiated Thu Apr 2 08:39:47 2026 as:
/usr/lib/nmap/nmap --privileged -sV -T4 -oN /home/kali/Desktop/nmap.txt scanme.nmap.org

Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.12s latency).

PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13
25/tcp    filtered  smtp
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
135/tcp    filtered  msrpc
139/tcp    filtered  netbios-ssn
179/tcp    filtered  bgp
445/tcp    filtered  microsoft-ds
646/tcp    filtered  ldap
9929/tcp   open      nping-echo   Nping echo
31337/tcp  open      tcpwrapped

Service Info: OS: Linux

# Nmap done: 1 IP address (1 host up) scanned in 98.35 seconds
```

Analysis

The scan revealed multiple open ports running network services. Open ports increase the attack surface of a system and should be monitored carefully.

Step 2 – Nikto Vulnerability Scan

Command Used

```
nikto -h scanme.nmap.org
```

Purpose

Nikto was used to detect:

- Outdated software

- Server misconfigurations
- Missing security headers
- Potential vulnerabilities

Scan Findings

The scan identified:

- Apache server information disclosure
- Missing security headers
- Apache mod_negotiation enabled
- Potential outdated Apache version

Results

```
- Nikto v2.6.0
-----
+ Target Host: scanme.nmap.org
+ Target Port: 80

+ GET /index: Uncommon header 'tcn' found.
+ Apache mod_negotiation is enabled with MultiViews.
+ Attackers may brute force file names.
+ Alternative file found: index.html

+ Apache/2.4.7 appears to be outdated
+ Current secure version is at least 2.4.66

+ Nikto scan completed
```

Analysis

These findings indicate possible security weaknesses in the web server configuration. Missing security headers may expose the website to attacks such as clickjacking and cross-site scripting (XSS).

Security Risks Identified

Vulnerability	Risk Level	Description
Open SSH Port	Medium	May allow brute-force login attempts
Outdated Apache Version	High	Could contain publicly known exploits
Missing Security Headers	Medium	Increases risk of web-based attacks

Vulnerability	Risk Level	Description
Exposed Services	Medium	Provides attackers with system information

Recommendations

- Update the Apache web server to the latest secure version
 - Implement proper HTTP security headers
 - Restrict unnecessary open ports
 - Configure firewall rules
 - Perform regular vulnerability scans
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Conclusion

This task demonstrated the use of reconnaissance and vulnerability scanning tools to identify open ports, running services, and potential security weaknesses in a target system. Proper security hardening and regular monitoring are essential to reduce cybersecurity risks.

Ethical Statement

All scanning activities were conducted on authorized and safe testing systems strictly for educational purposes only.